

Meeting 8

Lex Janssens

Date:	24 July 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
Mentor:	dr.ir. Á. Serra-Gomez
Project term:	February 2025 – August, 2025
E-mail:	l.p.janssens@umail.leidenuniv.nl

Outcome of previous meeting

- Discussed presenting the results (aligning axis, subsampling, final evaluation values in a table)
- Discussed possible publication opportunities, planned the first draft deadline, thesis defense deadline, and a loose final deadline.
- Discussed sharing the Overleaf for early reviewing
- Finalize the experiments, including TorchRL

Results during the last weeks

- Results are in. They are presented in aforementioned manner.
 - The performance of most algorithms under the hyperparameter configuration given by the original paper mostly matches, with a few exceptions of under- or overperformance on mainly Swimmer-v4, Walker2d-v4, and Ant-v4.
 - Comparability between baseline framework algorithm implementations often differs in specific environments: Swimmer-v4, Hopper-v4, and Walker2d-v4, where most inconsistent behaviour is found in PPO implementations.
 - The relative ranking between algorithms is mostly preserved. Using Stable Baselines 3 default hyperparameters, it differs in Swimmer-v4 and Walker2d-v4. Most differences stem from the fact that the algorithm performance differences are smaller compared to other frameworks.
- Overleaf is made public, and includes only complete parts of the thesis.
- Aske Plaat will be away from the 26th until the 24th of August, which is why I couldn't confirm the thesis defense date.

Plans for next weeks

- Finalize thesis defense details.
- Finish the first draft, which is still on schedule for the planned 1st of August.